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Supplementary Materials for

Familial autoinflammation with neutrophilic dermatosis reveals a regulatory mechanism of pyrin activation

Seth L. Masters,* Vasiliki Lagou, Isabelle Jéru, Paul J. Baker, Lien Van Eyck, David A. Parry, Dylan Lawless, Dominic De Nardo, Josselyn E. Garcia-Perez, Laura F. Dagley, Caroline L. Holley, James Dooley, Fiona Moghaddas, Emanuela Pasciuto, Pierre-Yves Jeandel, Raf Sciot, Dena Lyras, Andrew I. Webb, Sandra E. Nicholson, Lien De Somer, Erika van Nieuwenhove, Julia Ruuth-Praz, Bruno Copin, Emmanuelle Cochet, Myrna Medlej-Hashim, Andre Megarbane, Kate Schroder, Sinisa Savic, An Goris, Serge Amselem, Carine Wouters,* Adrian Liston*

*Corresponding author. E-mail: masters@wehi.edu.au (S.L.M.); carine.wouters@uzleuven.be (C.W.); adrian.liston@vib.be (A.L.)

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Other Supplementary Material for this manuscript includes the following:
(available at www.sciencetranslationalmedicine.org/cgi/content/full/8/332/332ra45/DC1)

Source data (Excel files)

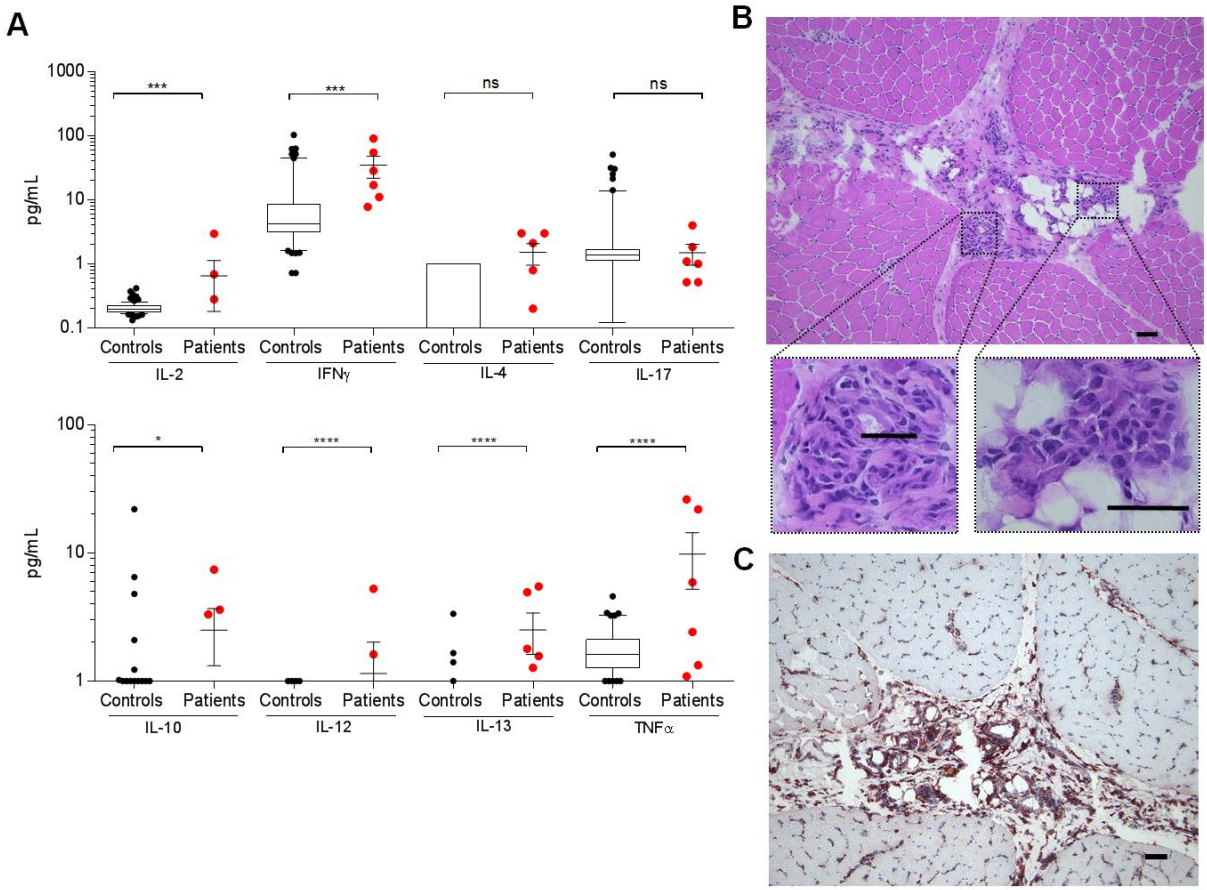


Figure S1. Inflammatory manifestations of PAAND. (a) Serum levels of IL-2, IFNγ, IL-4, IL-17, IL-10, IL-12, IL-13 and TNFα in patients during inflammatory episodes (n=6) and healthy controls (n=121). For healthy controls box and whiskers represent 25-50% and 5-95% range. For patients, mean and standard deviation shown. ***, p<0.001; ****, p<0.0001. **(b)** Histologic presentation of inflammatory infiltrate in the skeletal muscle of patient III.3. Mononuclear infiltrate in the perimysium by H&E (bar = 50µm) is shown at higher magnification. **(c)** Macrophages were assessed through CD68 immunohistochemistry in the skeletal muscle of patient III.3 (bar = 50µm).

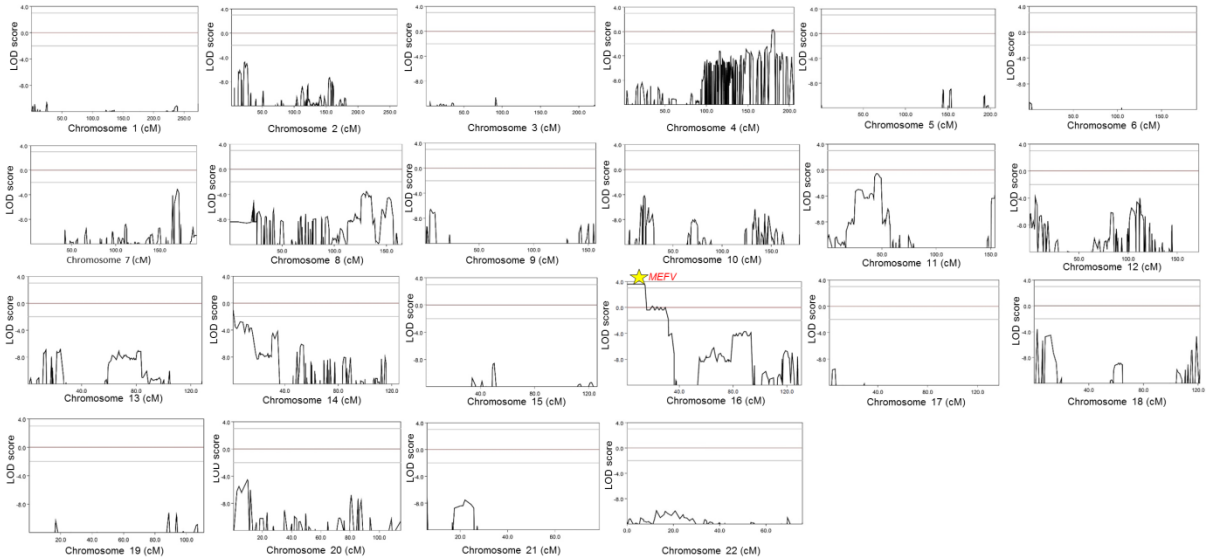


Figure S2. Results from linkage analysis. Genome-wide analysis for 20 family members (12 affected, 10 unaffected), indicating LOD scores for each chromosome.

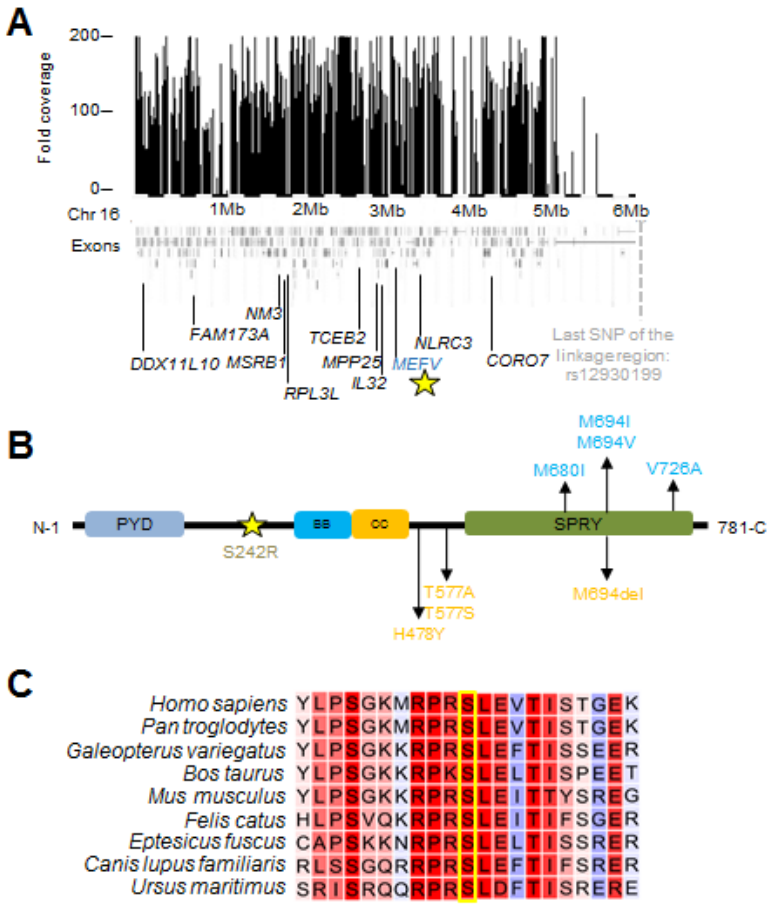


Figure S3. Dominant inheritance of PAAND caused by mutation in pyrin, S242R. (a) The gene-structure of the linkage (Chr16:1-6,283,321), indicating average fold coverage in patients and genes with differential expression in peripheral blood leukocytes. Exome sequencing in this region identified a single synonymous variant in RPL3L, and a single missense mutation in MEFV. **(b)** Domain structure of pyrin, indicating the locations of the observed S242R mutation (yellow star), known autosomal dominant mutations causing FMF (orange) and known autosomal recessive mutations causing FMF (blue). **(c)** Cross-species conservation of pyrin in the region flanking S242. The background colour indicates the level of conservation (red for highly conserved positions and blue for less conserved).

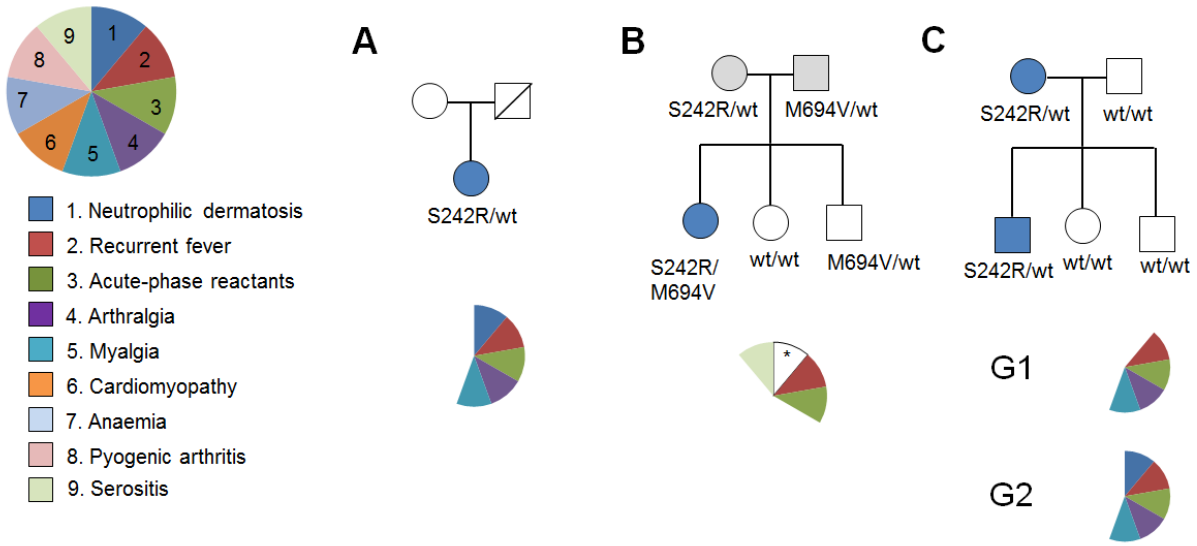


Figure S4. Inheritance of S242R mutations in additional pedigrees. (a) Known pedigree and clinical condition of a French family carrying the S242R mutation. Parental testing was not available to determine whether the mutation was inherited or de novo. **(b)** Known pedigree and clinical condition of a Lebanese family carrying the S242R mutation. Parental testing revealed that this was a compound heterozygous allele. Gray indicates clinical testing was not possible. The mother self-reported as healthy. **(c)** Known pedigree and clinical condition of a British family carrying the S242R mutation. * transient skin rashes/nodules.

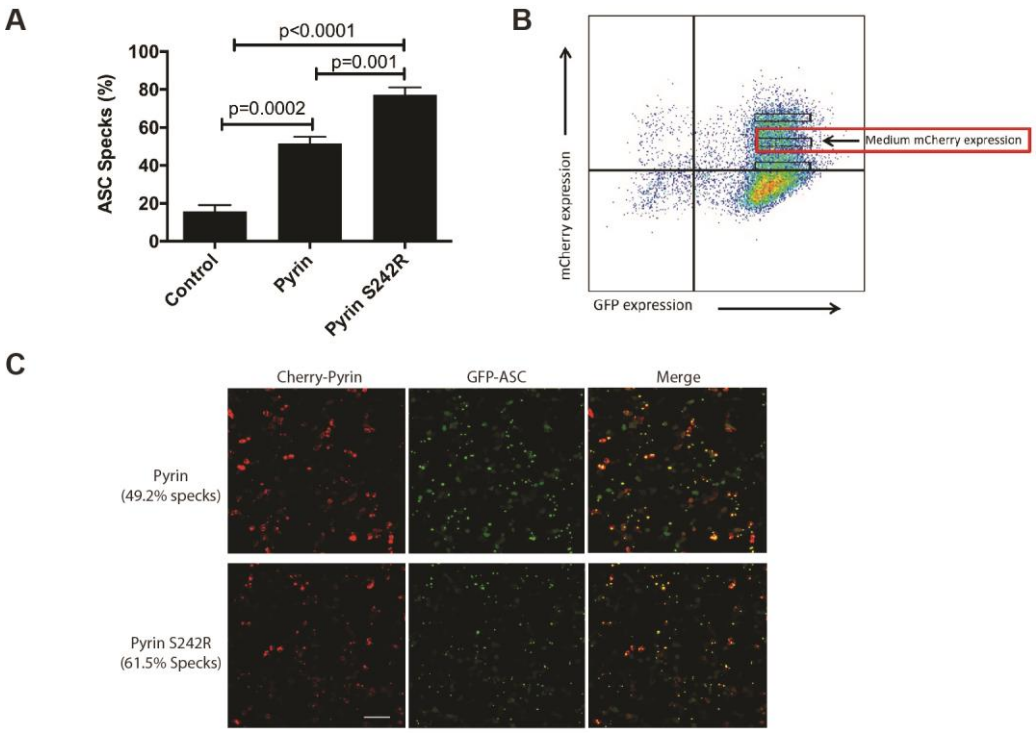


Figure S5. Fluorescent microscopy of pyrin/ASC specks. (a) Quantification of three biological replicates of ASC specks as a percentage of transfected (GFP +ve) cells, as in Figure 2a. **(b)** Gating strategy for cells expressing a similar (moderate) level of pyrin (mCherry) via FACS, for subsequent analysis of GFP specs in Figure 2b. **(c)** Colocalization of mCherry-Pyrin and GFP-ASC, as determined by confocal microscopy, with mCherry-Pyrin specks quantified. Scale bar = 100um. Representative of at least 3 biologically-independent experiments. Error bars represent mean + SD for n=3 with t-test.

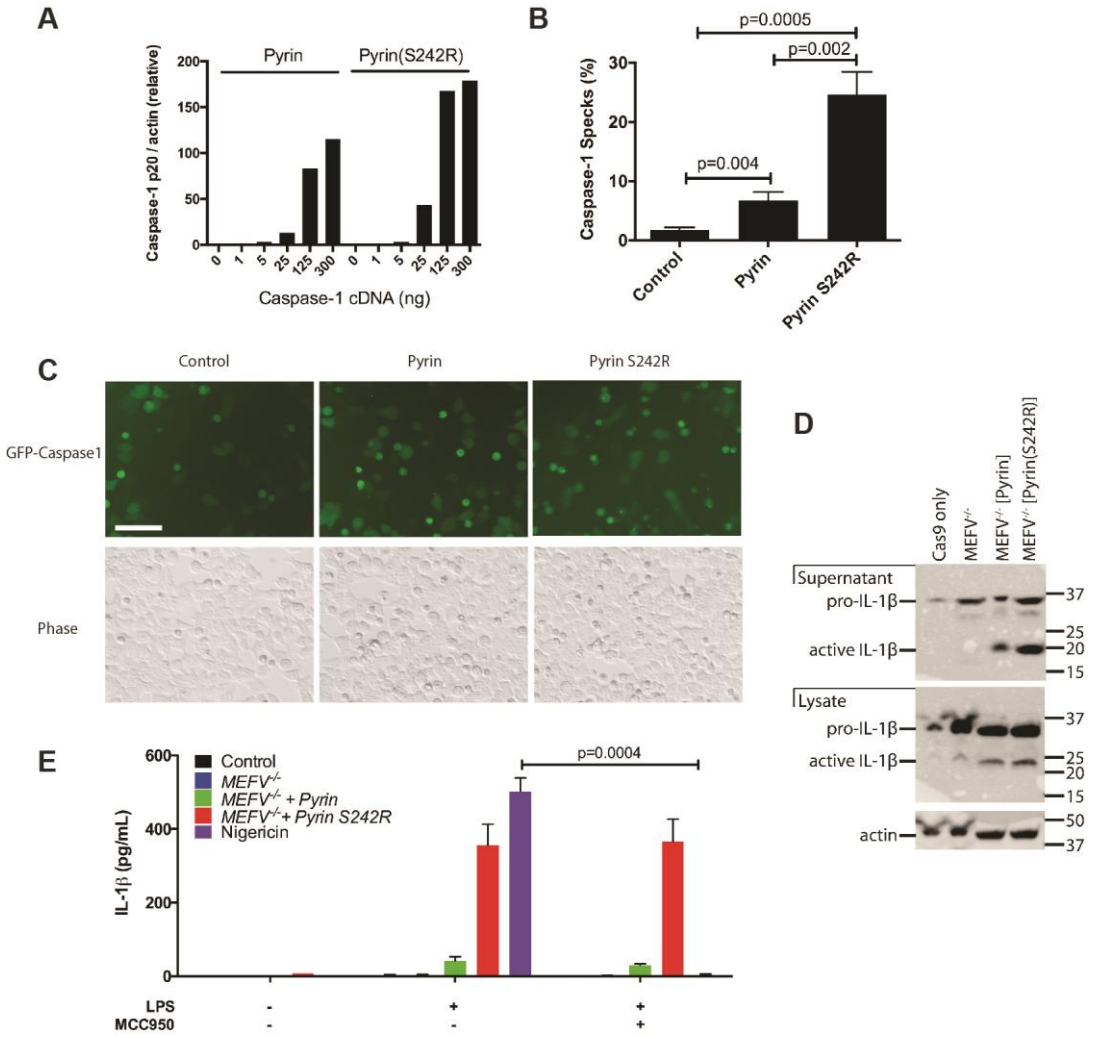
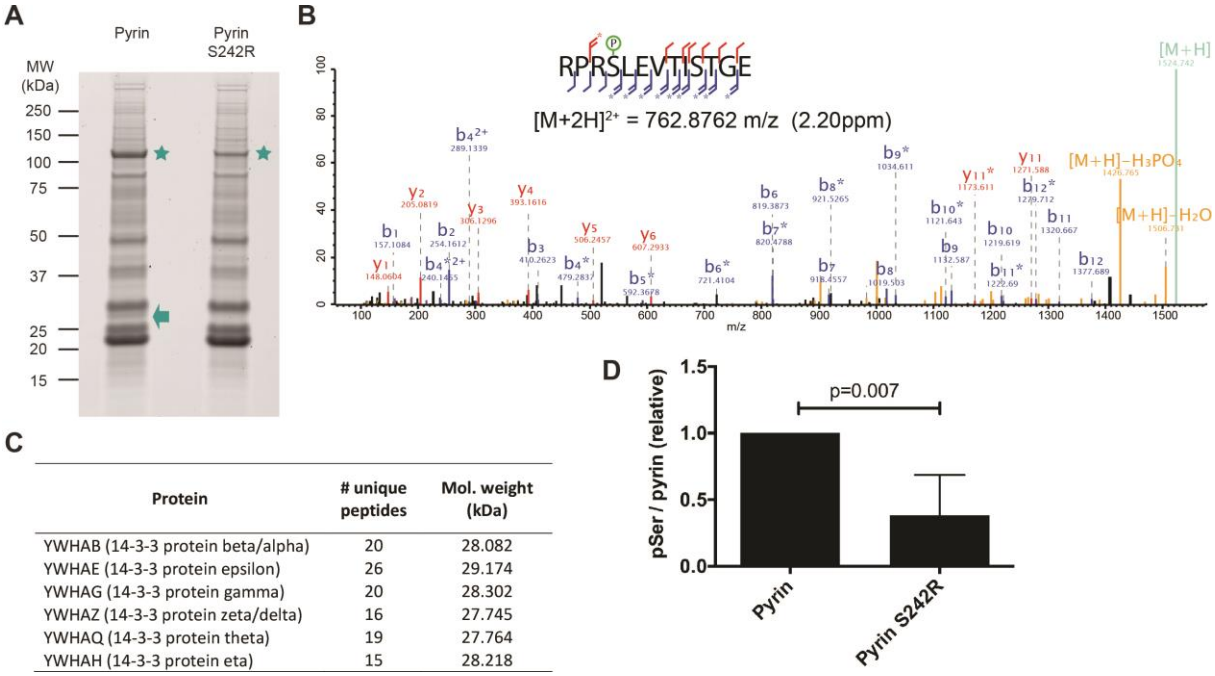


Figure S6. Activation of caspase-1 and IL-1β by pyrin S242R. **(a)** Quantification of cleaved caspase-1 p20 fragment relative to actin in the western blot Figure 2d. **(b)** Confocal microscopy showing a pyrin-driven increase in recruitment of GFP-Caspase-1 to cytoplasmic specks of ASC when co-transfected into HEK293T cells with wildtype or mutated pyrin (S242R). Scale bar = 50μm **(c)** Quantification of three biological replicates of Caspase-1 specks as a percentage of transfected (GFP positive) cells, as in panel (b). **(d)** Western blot for IL-1β, in supernatants and lysates of control (Cas9 only) and MEFV$^{-/-}$ THP-1 cells reconstituted with wild-type or S242R pyrin and treated with LPS for 24 hrs**. (e)** Monocytic THP-1 cells used in Figure 2e/f were additionally stimulated with 1 μM nigericin for 1 hr, with or without 1 μM MCC950, then IL-1β was measured by ELISA. Experiments

representative of at least 3 biologically-independent experiments. Error bars represent mean + SD for n=3 with t-test.



Protein	# unique peptides	Mol. weight (kDa)
YWHAB (14-3-3 protein beta/alpha)	20	28.082
YWHAE (14-3-3 protein epsilon)	26	29.174
YWHAG (14-3-3 protein gamma)	20	28.302
YWHAZ (14-3-3 protein zeta/delta)	16	27.745
YWHAQ (14-3-3 protein theta)	19	27.764
YWHAH (14-3-3 protein eta)	15	28.218

Figure S7. Proteomic analysis of pyrin interactions and phosphorylation. (a) GST-tagged pyrin with or without the S242R variant was overexpressed in HEK293T cells, then immunoprecipitated using glutathione sepharose 4B, with eluates run on a SDS-PAGE gel and stained with Sypro Ruby followed by Coomassie G-250. Stars indicate the bands corresponding to pyrin or pyrin S242R, while the band bound specifically to pyrin, indicated by an arrow, was excised for proteomic analysis. **(b)** Summary of the proteomic analysis for the band marked by an arrow on the SDS-PAGE gel in panel (a). **(c)** MS/MS spectrum identifying pyrin phosphorylation site at S242. Stars indicate neutral loss fragmentation ions, clearly demonstrating phosphate localisation at serine 242. Representative of at least 3 biologically-independent experiments. Error bars represent mean + SD for n=5 with t-test.

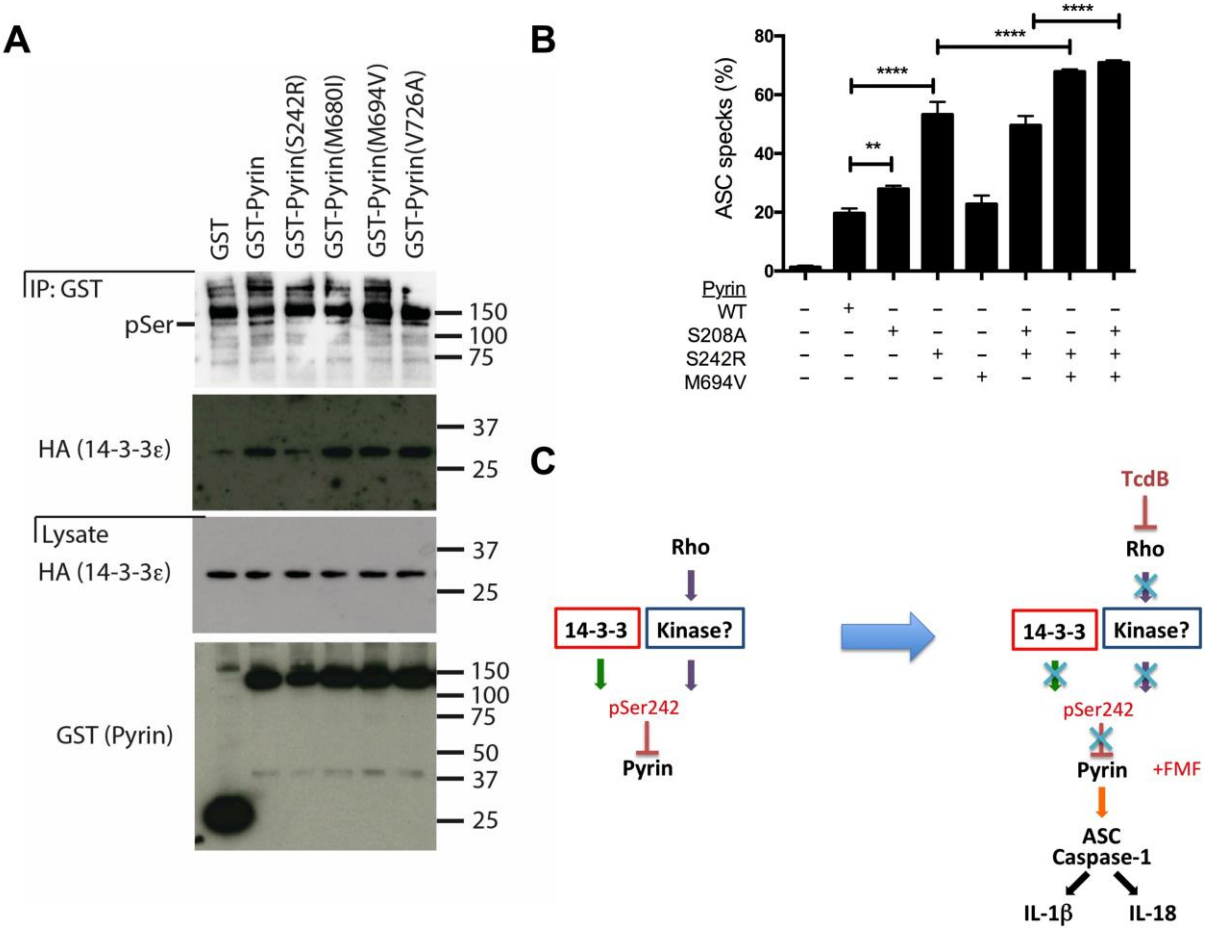
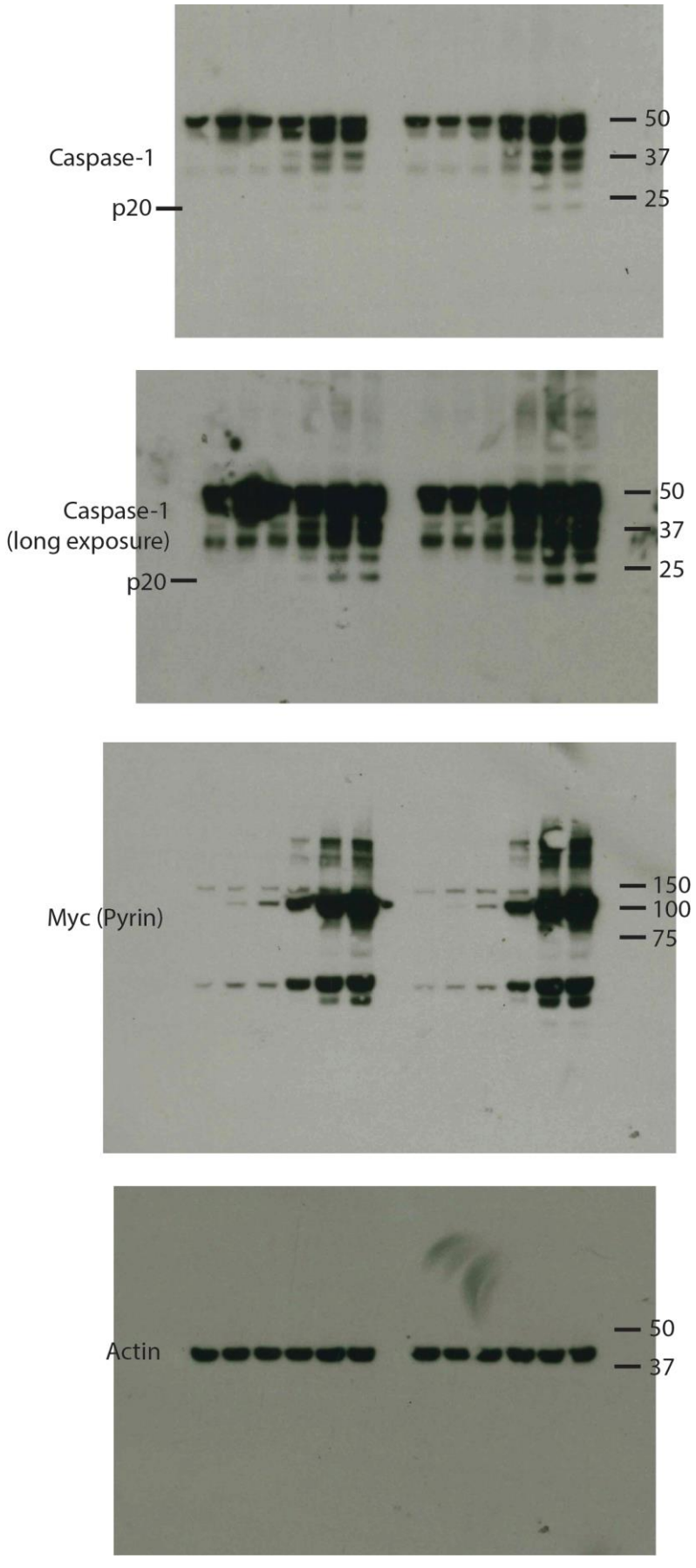
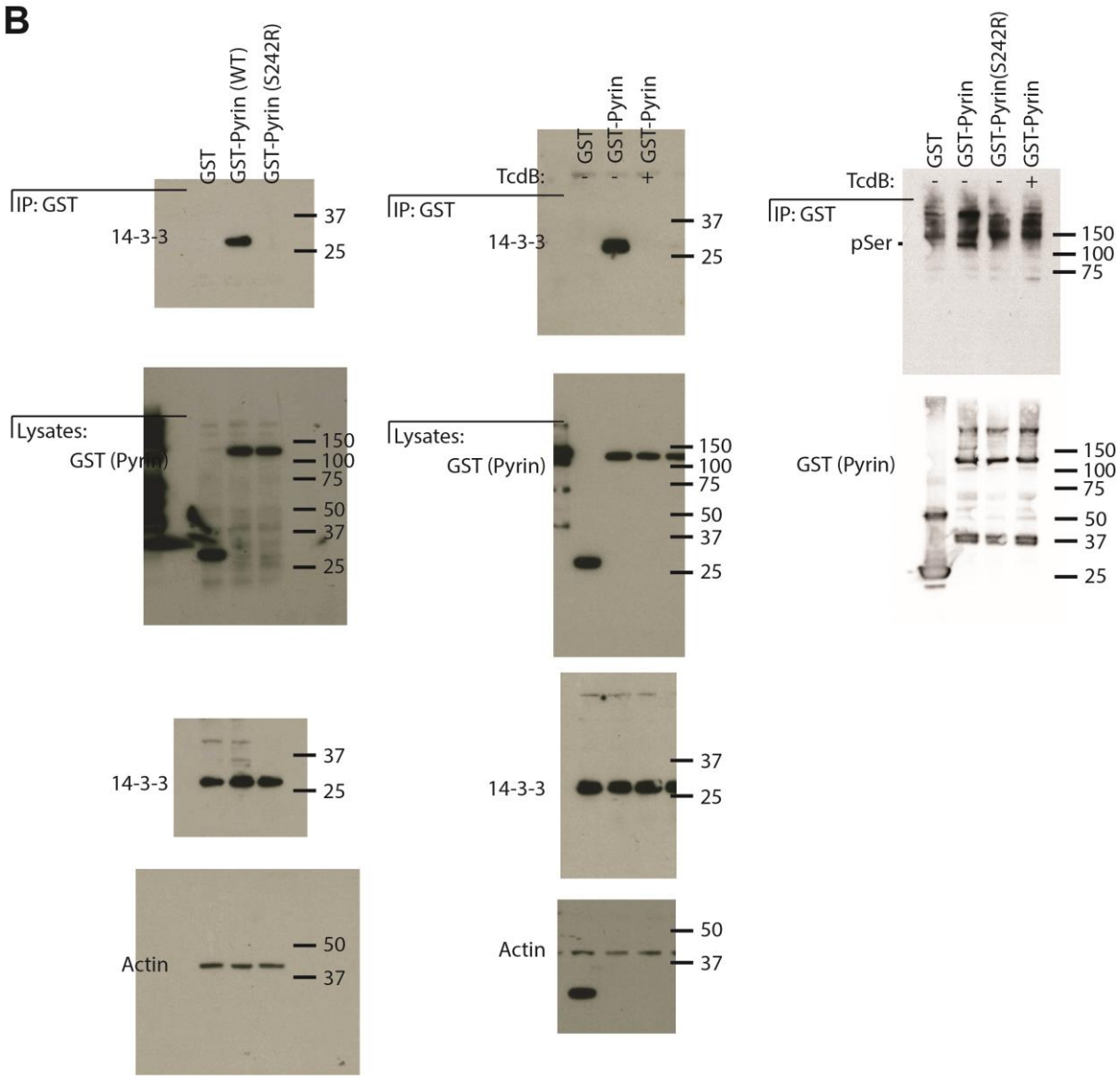
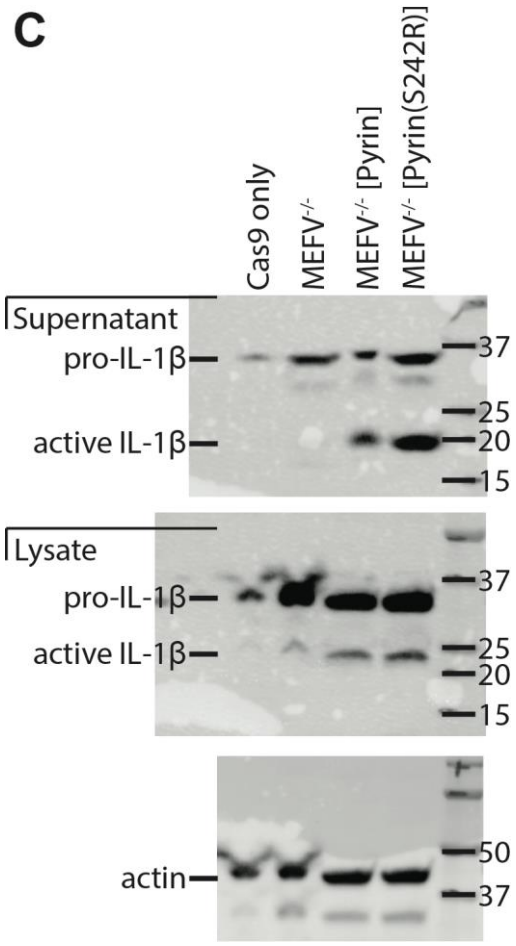


Figure S8. Functional effect of FMF mutations in context of pyrin S242R. (a) GST, or GST-Pyrin variants were expressed in HEK293T cells, together with HA-14-3-3ε. Lysates or eluates from GST immunoprecipitation (IP) were subjected to western blotting for pSer (14-3-3 motif), pyrin (GST) or 14-3-3ε (HA). **(b)** Quantification of 3 biological replicates for GFP-ASC specks by flow cytometry as in Figure 2, with the overexpression of different pyrin mutants. **(c)** Schematic for the mechanism of pyrin activation. All representative of at least 3 biologically-independent experiments. Error bars represent mean + SD for n=3 with t-test.

A


Caspase-1
p20
50
37
25
Caspase-1
(long exposure)
p20
50
37
25
Myc (Pyrin)
150
100
75
Actin
50
37


B
GST
GST-Pyrin (WT)
GST-Pyrin (S242R)
IP: GST
14-3-3
37
25
Lysates:
GST (Pyrin)
150
100
75
50
37
25
14-3-3
37
25
Actin
50
37
TcdB:
GST
GST-Pyrin
GST-Pyrin
-
-
+
IP: GST
14-3-3
37
25
Lysates:
GST (Pyrin)
150
100
75
50
37
25
14-3-3
37
25
Actin
50
37
TcdB:
GST
GST-Pyrin
GST-Pyrin(S242R)
GST-Pyrin
-
-
-
+
IP: GST
pSer
150
100
75
GST (Pyrin)
150
100
75
50
37
25


C
Cas9 only
MEFV-/-
MEFV-/- [Pyrin]
MEFV-/- [Pyrin(S242R)]
Supernatant
pro-IL-1β
active IL-1β
37
25
20
15
Lysate
pro-IL-1β
active IL-1β
37
25
20
15
actin
50
37

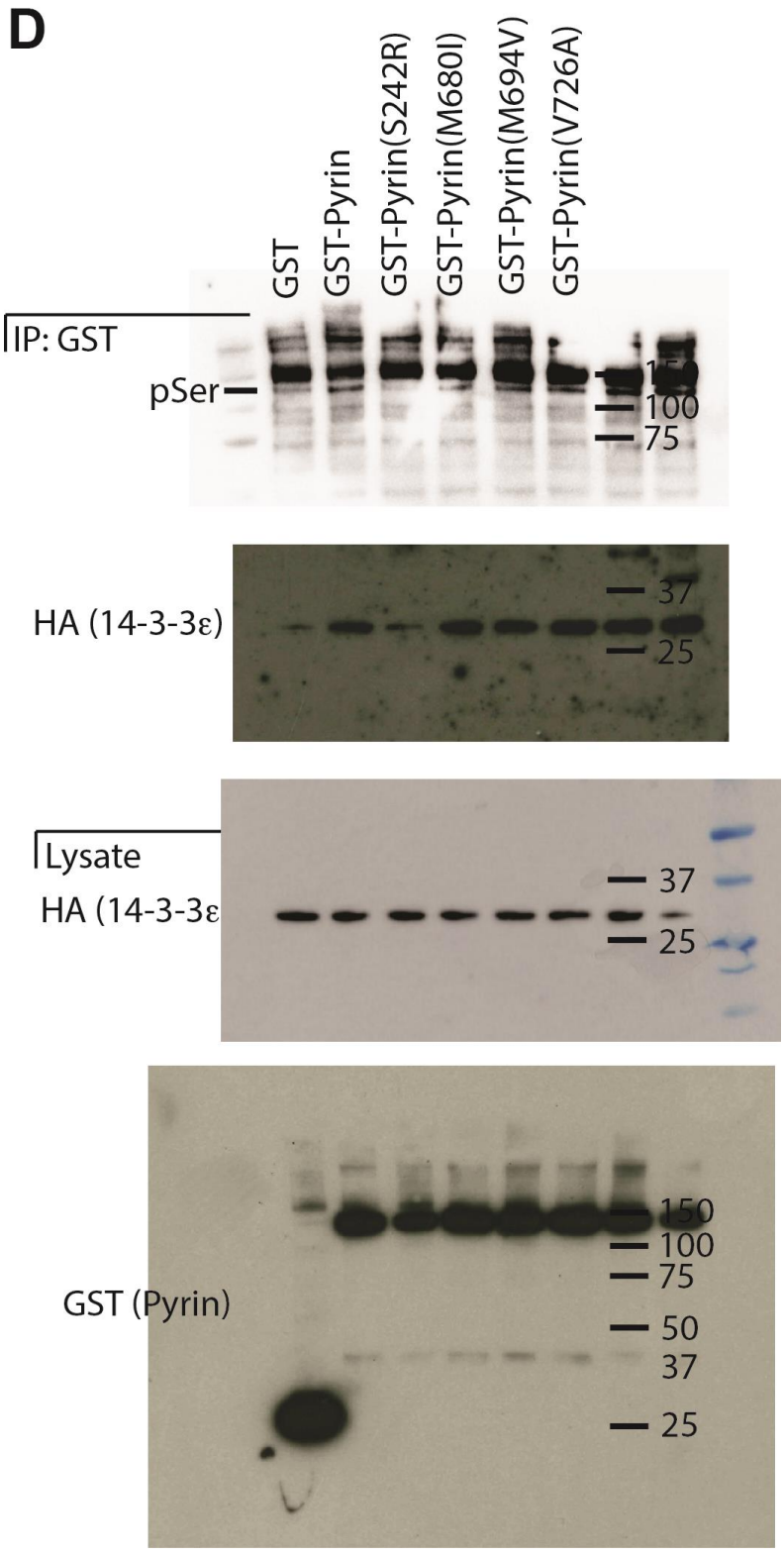


Figure S9. Full Western blots. (a) Full western blots from Figure 2d. **(b)** Full western blots from Figure 3a,b,c **(c)** Full western blots from Figure S6d. **(d)** Full western blots from Figure 8a.